

MPV DOWNLOAD, SETUP AND CUSTOMISATION GUIDE

Windows setup, from a clean download to a fully customised ModernZ-based player

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Player	mpv (media player, Windows build)
Source build	zhongfly/mpv-winbuild
OSC used	ModernZ (fork of ModernX)
Config link	https://github.com/Biraj2004/biraj-mpv-conf

1 What is mpv, and why use it

mpv is a free, open-source media player for multiple platforms, built on MPlayer and mplayer2. Unlike VLC, it does not come with a heavy built-in interface. Instead, it is designed to be lightweight, scriptable, and easy to customize using plain text files and Lua scripts. This allows you to build a player that looks and works exactly how you want, rather than being restricted to a fixed design.

Because of this, a new mpv installation has almost no visible controls on the screen. Everything in this guide — the interface, seek bar, thumbnail previews, subtitles, and keyboard shortcuts — was added step by step. This document explains exactly what to download, where to put each file, and what every setting does.

2 Downloading mpv for Windows

2.1 Starting point: the official site

The official mpv website is <https://mpv.io/installation/>. Windows does not have an official installer made by the mpv team, so the installation page points users to Windows builds maintained by the community.

2.2 Build actually used: mpv-winbuild

From the suggested links, the specific build downloaded was:

<https://github.com/zhongfly/mpv-winbuild/releases>

This page regularly compiles mpv for Windows and provides ready-to-use `.7z` archives containing `mpv.exe` and the required DLL files. The steps followed were:

- Open the releases page above.
- Download the latest release archive (a `.7z` file).
- Extract the archive using 7-Zip and place the contents into the permanent folder `C:\Program Files\mpv\`.
- Inside this folder, you will find items such as the `doc`, `installer`, `mpv`, and `mpv-cache` folders, along with files like `mpv.com`, `mpv.exe`, `settings.xml`, and several batch scripts.
- Right-click on the `mpv-register.bat` and `mpv-updater.bat` scripts and select **Run as Administrator**. Doing this will set up the required internal settings and automatically add mpv to your Windows `PATH` environment variable, so it can be easily opened from the Command Prompt or PowerShell.

mpv on Windows is portable. It does not need a standard installer. The extracted folder is the full application. All your personal settings (scripts, themes, key bindings) are kept safely separate in your user `AppData` folder, which is explained next.

3 Where mpv reads its configuration from

mpv looks for a specific configuration folder inside your Windows roaming application data directory. For the user `biraj`, this path is:

```
C:\Users\biraj\AppData\Roaming\mpv\
```

This folder does not exist by default. You have to create it manually the first time. Every configuration file discussed in this guide belongs inside it. mpv automatically checks this folder every time it opens, so as long as the folder and file names match exactly what mpv expects, no extra setup is needed.

3.1 Final folder structure used

```
mpv/  
  fonts/  
    modernz-icons.ttf  
  script-opts/  
    modernz.conf  
    thumbfast.conf  
    pause_indicator_lite.conf
```

```
scripts/
  modernz.lua
  thumbfast.lua
  open-file.lua
  pause_indicator_lite.lua
input.conf
menu.conf
mpv.conf
```

mpv is very strict about these exact names. The folder must be named **script-opts** (singular "script"), not **scripts-opts**. The main configuration file must be named **mpv.conf**, not **mpv.config**. If these are wrong, your settings will be silently ignored without showing any errors.

4 ModernZ – the on-screen controller

By default, mpv shows almost no interface. ModernZ is a third-party Lua script that replaces the default screen with a modern, skinnable on-screen controller (OSC). It provides a seek bar, play/pause buttons, volume control, subtitle menus, chapter markers, and a screenshot button.

<https://github.com/Samillion/ModernZ>

4.1 Files it requires

File	Purpose	Placed in
modernz.lua	The OSC script itself	scripts\
modernz.conf	Layout, colour, and behaviour settings	script-opts\
modernz-icons.ttf	Icon font used for the buttons	fonts\

4.2 Base requirement in mpv.conf

ModernZ replaces mpv's native OSC completely, so the built-in one must be turned off first:

```
osc=no
title-bar=no
```

5 Supporting scripts actually used

Alongside ModernZ, three additional scripts were added from ModernZ's own "extras" collection, and one separate script for thumbnail previews.

5.1 thumbfast – live thumbnail previews

<https://github.com/po5/thumbfast>

This generates a live thumbnail image when hovering over the seek bar, showing what the video looks like at that specific time. Placed as `scripts\thumbfast.lua`, and configured through `script-opts\thumbfast.conf`.

5.2 open-file.lua – native Windows file dialog

<https://github.com/Samillion/ModernZ/tree/main/extras/open-file>

This adds the ability to open files, external subtitle tracks, and external audio tracks using the standard Windows file picker dialog. It is bound to `Ctrl+O`, `Ctrl+S`, and `Ctrl+A` respectively.

5.3 pause_indicator_lite.lua – pause and mute indicator

<https://github.com/Samillion/ModernZ/tree/main/extras/pause-indicator-lite>

Displays a small icon on the screen whenever playback is paused, unpaused, or muted. It is styled to match the icon theme of the main OSC. Configured through `script-opts\pause_indicator_lite.conf`.

6 Core mpv.conf settings used

The main settings file is placed at `C:\Users\biraj\AppData\Roaming\mpv\mpv.conf`. It controls playback behaviour, video and audio quality, and subtitle handling.

6.1 General window and OSC behaviour

```
osc=no
title-bar=no
keep-open=yes

autofit=80%x80%
autofit-larger=80%x80%
autofit-smaller=640x360
geometry=50%:50%
```

```
cursor-autohide=1000
cursor-autohide-fs-only=yes

# Prevents mpv's native seek bar overlay from
# doubling up with the ModernZ seek bar
osd-bar=no
```

6.2 Resume playback and watch history

```
save-position-on-quit=yes
save-watch-history=yes
watch-later-options=start,vid,aid,sid,volume,mute
watch-later-options-remove=sub-pos
watch-later-options-remove=osd-margin-y
```

save-position-on-quit remembers the exact time a file was closed and resumes from there the next time it is opened. **save-watch-history** keeps a log of recently played files. The **remove** lines stop mpv from saving subtitle position values, which previously caused subtitles to behave strangely after resuming a video.

6.3 Video quality

```
vo=gpu-next
hwdec=d3d11va

deband=yes
temporal-dither=yes

sigmoid-upscaling=yes
correct-downscaling=yes
```

hwdec=d3d11va turns on hardware-accelerated decoding, which lowers CPU usage. **deband** smooths out blocky colours often visible in dark scenes. The upscaling options improve how mpv resizes the video on your screen.

6.4 Audio

```
volume=70
volume-max=150
audio-file-auto=fuzzy
audio-pitch-correction=yes
```

6.5 Subtitles

```
sub-auto=yes
blend-subtitles=yes
sid=no

demuxer-mkv-subtitle-preroll=yes
demuxer-mkv-subtitle-preroll-secs=2

embeddedfonts=yes
sub-fix-timing=no
sub-use-margins=yes
sub-ass-force-margins=yes

sub-font-size=30
sub-outline-size=0
sub-shadow-offset=4
sub-back-color=0/0.6
sub-border-style=background-box
sub-file-paths=sub;subs;subtitles;srt;ass
```

Combining **sub-auto=yes** with **sid=no** means mpv will find and load any matching subtitle files automatically, but it will not display them until you choose to. The font styling settings only apply to basic text subtitles (**.srt** or **.vtt**). Advanced subtitle files that carry their own built-in styling will render exactly as their creators intended.

6.6 Languages

```
slang=en,enm,hi
alang=hi,en,ja
```

6.7 Screenshots

```
screenshot-format=png
screenshot-png-compression=6
screenshot-directory=~ / Pictures / mpv-screenshots
screenshot-template=%F- (%P) - %n
```

6.8 Cache and yt-dlp integration

```
cache=yes
demuxer-max-bytes=400MiB
demuxer-max-back-bytes=200MiB

ytdl=yes
script-opt=ytdl_hook-all_formats=yes
```

This allows mpv to directly open streaming URLs (like YouTube) using yt-dlp to load the actual media stream.

6.9 Situational profiles

These blocks apply extra settings only when a specific condition is met, without affecting normal playback.

Profile	Effect
Window-PiP	When the window is pinned on top and not fullscreen, it scales the OSC up slightly and keeps the progress bar visible.
Minimized	Automatically pauses playback the moment the window is minimised.
Video	Enables taskbar progress display specifically for video files.
Image	When an image is opened, it disables taskbar progress, keeps the image displayed indefinitely, and re-centres it.

7 ModernZ configuration (modernz.conf)

Located at `script-opts\modernz.conf`. Below are the specific options changed from their defaults to make the player cleaner.

Option	Set to	Reason
<code>show_title</code>	<code>no</code>	Hides the filename text shown on the OSC.
<code>osc_on_seek</code>	<code>no</code>	Stops the main bar from popping up on the screen every time you skip forward or backward.
<code>deadzonesize</code>	<code>0</code>	Originally <code>0.75</code> . Set to <code>0</code> so moving the mouse anywhere in the window reveals the controls.
<code>minmousemove</code>	<code>0</code>	Any mouse movement, however small, triggers the controls.

8 Thumbnail previews (thumbfast.conf)

Located at `script-opts\thumbfast.conf`. The thumbnail images generated by this script are cached directly in the `mpv-cache` folder found within the main `C:\Program Files\mpv` installation directory.

Option	Reason
<code>thumbnail</code>	Sets the cache path to <code>C:\Program Files\mpv\mpv-cache\Thumbnail</code> to store generated preview files safely.
<code>hwdec=no</code>	Forces software decoding for the background thumbnail process to prevent solid black preview boxes.
<code>spawn_first=yes</code>	Starts the background thumbnail process as soon as a video loads.
<code>max_height</code> / <code>max_width</code>	Caps thumbnail resolution to <code>200</code> to keep generation fast.
<code>overlay=yes</code>	Renders thumbnails directly to the screen, which is much faster than saving image files to your disk.

9 Pause and mute indicator (pause_indicator_lite.conf)

Located at `script-opts\pause_indicator_lite.conf`. Only the changed values are listed here.

Option	Effect
<code>indicator_stay=no</code>	The pause icon fades out after a short timeout instead of staying on screen forever.
<code>icon_opacity=35</code>	Keeps the icon subtle and slightly transparent.
<code>themed_icons=yes</code>	Uses the ModernZ font icons instead of basic drawn shapes.
<code>mute_indicator=yes</code>	Adds a small speaker-mute icon in the corner whenever audio is muted.

10 Keyboard shortcuts (input.conf)

Located at `C:\Users\biraj\AppData\Roaming\mpv\input.conf`. This file was trimmed down to essential bindings only, styled closely to VLC's layout.

Key	Action
Space	Play / pause
Left / Right arrow	Seek 2 seconds back / forward
Up / Down arrow	Volume up / down
b	Cycle audio track (VLC-style)
v	Cycle subtitle track (VLC-style)
m	Mute / unmute
f	Toggle fullscreen
n / p	Next / previous file in playlist
. / ,	Frame step forward / backward
Tab	Toggle OSC visibility
Ctrl+O	Open file (native Windows dialog)
Ctrl+S / Ctrl+A	Add external subtitle / audio file

10.1 Why arrow-key seeking uses keyframes

Seeking with **exact** forces mpv to decode frame by frame to land precisely on the requested timestamp. This briefly pauses internal playback, triggering the pause indicator to flash on every key press. Switching to **keyframes** makes the jump land on the nearest keyframe instead, which is instant and does not trigger the pause icon.

11 Summary of problems fixed during setup

Problem	Fix
Wrong folder/file names	Renamed them to the exact names mpv expects (<code>script-opts</code> and <code>mpv.conf</code>).
Double seek bar showing	Added <code>osd-bar=no</code> in <code>mpv.conf</code> to disable mpv's native seek bar.
Thumbnail preview showing solid black	Set <code>hwdec=no</code> inside <code>thumbfast.conf</code> .
Subtitles auto-displaying by default	Set <code>sid=no</code> in <code>mpv.conf</code> , alongside <code>sub-auto=yes</code> .
OSC popping up on arrow-key seek	Set <code>osc_on_seek=no</code> in <code>modernz.conf</code> .

12 Configuration repository

The complete, up-to-date set of configuration files described in this guide is published at:

<https://github.com/Biraj2004/biraj-mpv-conf>

To reuse this setup on a fresh Windows computer, download my configuration files from the GitHub repository and paste them exactly as they are into the mpv folder inside your Roaming directory. The full path where these files must be pasted is:

```
C:\Users\<username>\AppData\Roaming\mpv\
```

Simply replace the `<username>` portion with your actual Windows account name.